ModelIt!

Student Activity Pack

Why You Can't Slide Forever

The Hidden Energy Transfers That Stop Everything

Grade 7 | NGSS MS-PS3-5 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why does a ball eventually stop rolling across the floor?

2 Where does the energy go when a moving object stops?

3 Why do your hands get warm when you rub them together?

4 Would an object ever stop moving in outer space? Why or why not?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Initial Speed | Surface Friction | Kinetic Energy | Thermal Energy Output

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Ice (Low Friction)

Very low surface friction, moderate initial speed

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Concrete (High Friction)

High surface friction, same initial speed

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Speed Comparison

Same surface, but double the initial speed

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Air Resistance — The friction force between a moving object and the air, which increases with speed

Mass — Heavier objects have more kinetic energy at the same speed and take more force to stop

Surface Area — More contact area between surfaces can increase the total friction force

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does every moving thing eventually stop — and where does the energy go? Use evidence from your simulation to support your answers.

1. **If energy can't be created or destroyed, where does ALL the energy from a moving car go when it brakes?**

2. **Why do space probes keep moving for decades without engines — doesn't that violate your model?**

3. **How is the friction that stops a skateboard the same concept as the friction that starts a campfire?**

4. **Why does doubling your speed quadruple your kinetic energy instead of just doubling it?**

5. **How does understanding energy transfer help engineers design better brakes?**

8 STEM Challenge: Design the Longest Slide

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YOUR ENGINEERING MISSION

Engineer a slide or ramp system that keeps an object moving the longest distance possible by minimizing energy lost to friction.

THE SCENARIO

A toy company wants to design the world's longest toy car track. Your engineering team must maximize the distance a toy car travels by understanding and minimizing energy losses.

Engineering Design Thinking

1. WHAT SURFACE MATERIAL WILL YOU USE FOR YOUR TRACK AND WHY?
2. WHAT STARTING HEIGHT WILL GIVE YOUR CAR THE MOST KINETIC ENERGY?
3. HOW WILL YOU MEASURE THE DISTANCE TRAVELED AND ENERGY LOST?
4. WHAT MODIFICATIONS CAN YOU MAKE TO REDUCE FRICTION EVEN FURTHER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers design systems that manage energy transfer — from car brakes to roller coasters to industrial machinery. They earn $75,000–$125,000/year.